Q1

a)

$z = DTI - Stability$

A : $z = -2$ $\quad \hat{p}_A = \frac{1}{1+e^{2}} = 0.119$

B : $z = -1$ $\quad \hat{p}_B = \frac{1}{1+e} = 0.269$

C : $z = 0$ $\quad \hat{p}_C = 0.5$

D : $z = 1$ $\quad \hat{p}_D = \frac{1}{1.368} = 0.73$

E : $z = 2$ $\quad \hat{p}_E = \frac{1}{1.135} = 0.88$

b)

$l_A = -(1)\ln(1-0.119) = 0.127$

$l_B = -(1)\ln(1-0.269) = 0.313$

$l_C = -\ln(1-0.5) = 0.693$

$l_D = -\ln(0.73) = 0.315$

$l_E = -\ln(0.88) = 0.128$

c) Odd ratios DTI $= e^{1} = 2.72$

Odd ratios stability $= e^{-1} = 0.368$

increasing DTI and stability by 1 is like multiplying odds by 2.72 and 0.368 respectively.